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Latent antibiotic resistance genes are abundant, diverse, and mobile in human, animal, and environmental microbiomes

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Background

- Bacterial communities in humans, animals, and the external environment maintain a large collection of antibiotic resistance genes (ARGs).
- Few of these ARGs are well-characterized and thus *established* in existing resistance gene databases.
- The remaining *latent* ARGs are typically unknown and overlooked in most sequencing-based studies.
- Our view of the resistome and its diversity is therefore incomplete, which hampers our ability to assess risk for promotion and spread of yet undiscovered resistance determinants.

Aims

- Compute the abundance and diversity of established and latent ARGs in in a wide range of environments.
- Identify widespread latent and established ARGs in similar hosts or environment and across them.
- Find bacterial communities acting as hot-spots for mobilization of ARGs into pathogens and other environments.
- Contribute to the knowledge on the origin, evolution and selection of ARGs and asses the challenges that this problem represents to public health.

Results

- The latent ARGs were more abundant and diverse than established ARGs in all studied environments, over 10,000 metagenomic samples including the human- and animal-associated microbiomes.
- The pan-resistomes were dominated by latent ARGs while the core-resistome comprised both latent and established ARGs.
- Several latent ARGs were shared between environments and/or present in human pathogens.
- Context analysis of these genes showed that they were located on mobile genetic elements.

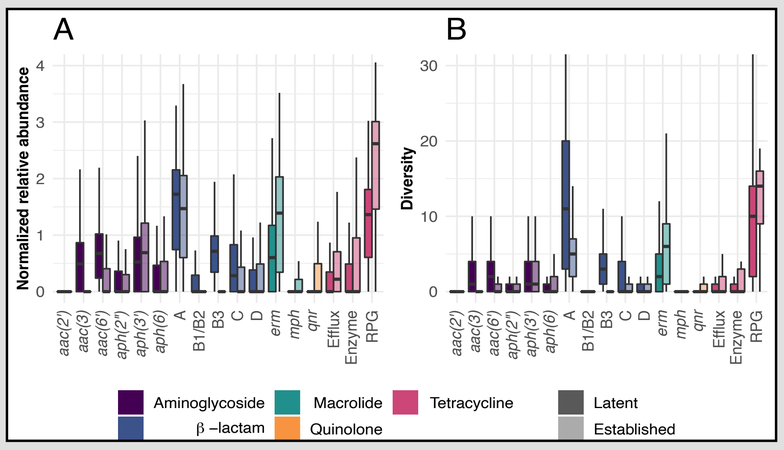

Figure 1. Distribution of the A) relative abundance and B) diversity for latent and established ARGs. Distribution of the log-transformed abundance and alpha-diversity per gene class.

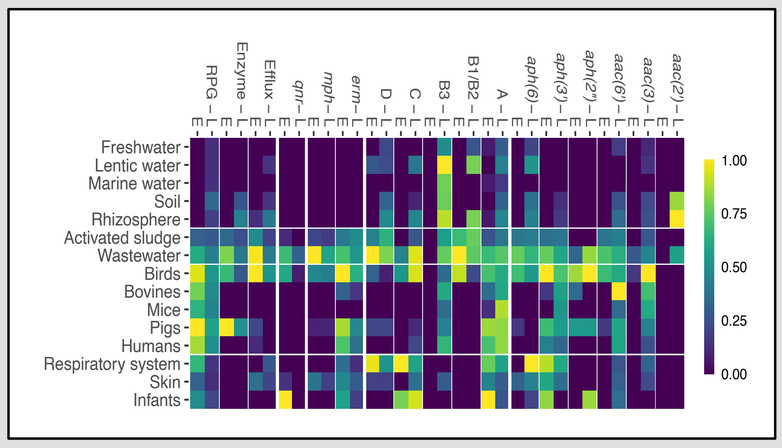

Figure 2. Relative abundance of latent and established ARGs per gene class and environment. Each gene class is represented by two columns: L for latent and E for established. The color intensity reflects the gene- and environment-specific relative abundance, which was calculated based on the median of the relative log-transform abundance over all samples from the environment. To make the genes comparable, all values were normalized based on the environment with the highest abundance.

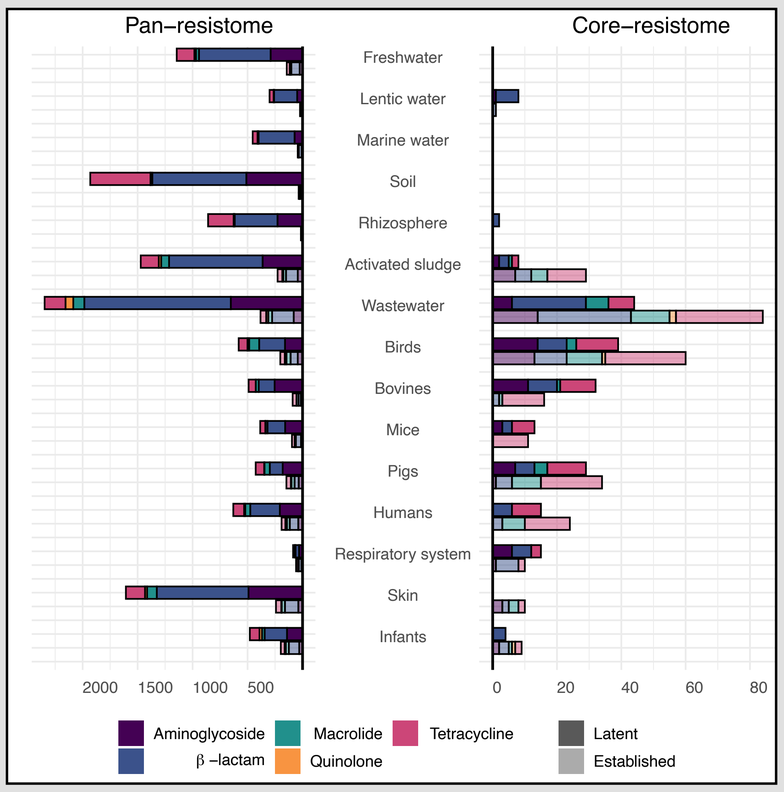

Figure 3. Pan-resistomes and core-resistomes. The pan resistome includes all genes encountered in at least one sample from the environment. The calculation for the core-resistome was done based on rarefied metagenomes that, for each environment, were repeatedly subsampled down to 100 samples. The core genes are those commonly encountered in at least 50 samples. The core-resistome shows the number of core genes present in 90% of the subsamples.

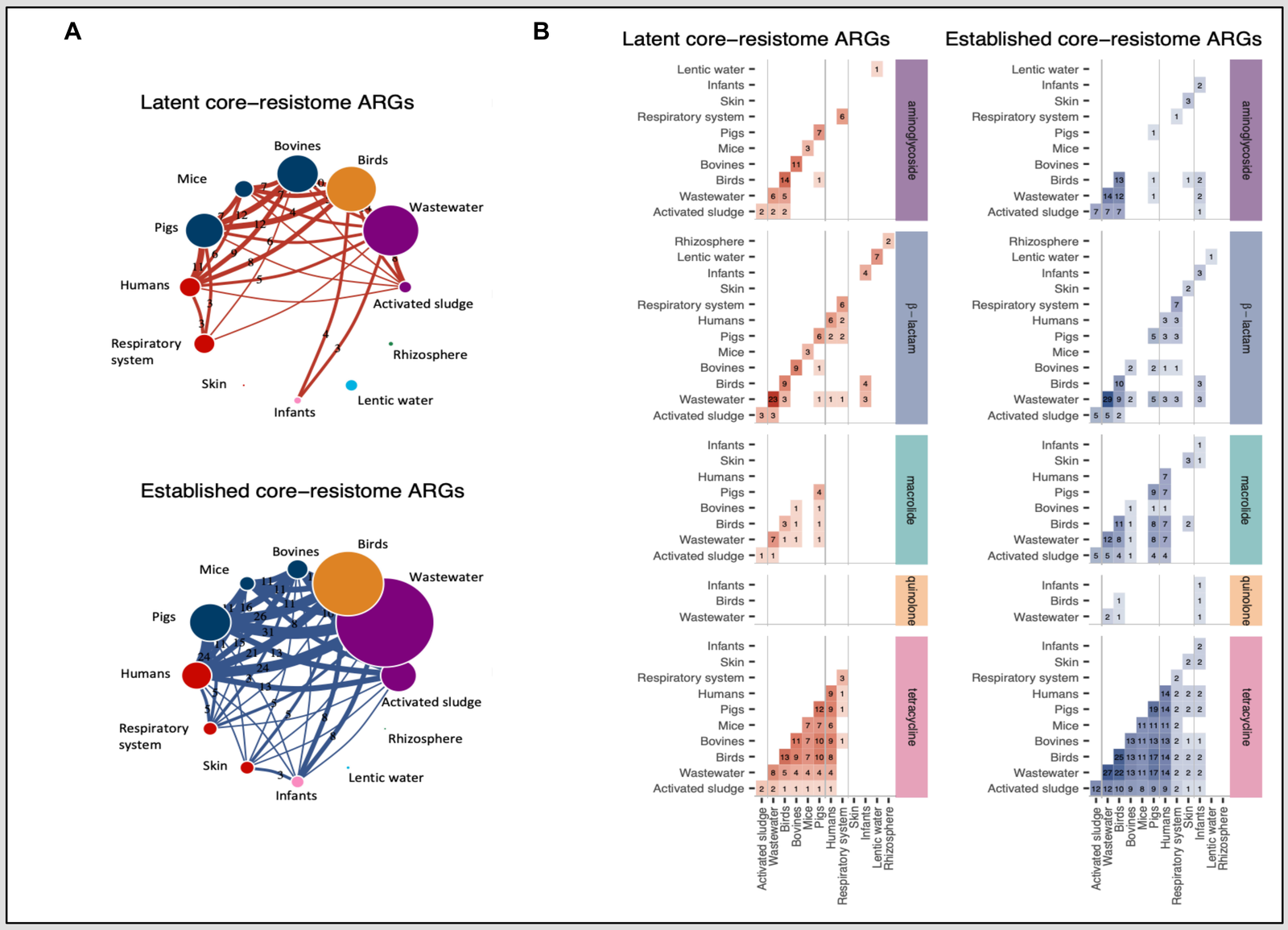

Figure 4. Core-resistome overlap between environments. **A)** Networks where nodes represent environments with their node size proportional to the size of the environment's core-resistome. The width of the edge is proportional to the number of core-resistome ARGs shared between two environments. Only overlaps of at least three ARGs are shown.

B) The heatmaps show the number of core-resistome ARGs shared between two environments for each antibiotic type.

Conclusions

- Latent ARGs are ubiquitously present in all environments and constitute a diverse reservoir from which new resistance determinants can be recruited to pathogens.
- Several latent ARGs already had high mobile potential and were present in human pathogens, suggesting that they may constitute emerging threats to human health.
- Wastewater microbiomes have a high potential for the mobilization and promotion of latent ARGs.
- The full resistome needs to be considered to properly assess the risks associated with antibiotic selection pressures.

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